

CHAPTER 2

LITERATURE STUDY

This chapter provides a comprehensive literature survey on WQM and WQP in aquaculture. It emphasizes the significance of WQM and WQP in mitigating environmental effects and optimizing fish growth. This literature survey explains WQM and various prediction models highlighting the limitations of traditional approaches and the potential of DL techniques. Attention mechanisms and hybrid models are discussed for enhancing prediction accuracy. The survey concludes by emphasizing the importance of accurate WQP in high-intensity aquaculture and the potential of attention-based DL models for improved WQP performance. Section 2.1 of this chapter provides literature survey on WQM and Section 2.2 provides literature survey on WQP. Section 2.3 give details on the datasets. The performance metrics are detailed in Section 2.4.

2.1 WATER QUALITY MONITORING IN AQUACULTURE

Aquaculture is evolving towards intensive controlled environment production, giving a substantial improvement in production. This WQM system in [8] has a set of sensors, gateways and computer. Sensors are used to monitor WQ parameters and water level, effectively reducing the risk of mortality by ensuring continuous WQ monitoring. The system can detect and control temperature, dissolved oxygen (DO) content, pH value and WQ parameters in real-time. Authors used LabView to display information and ZigBee for communication. This work has an extensive study for six months with a study on battery and communication performance. This [21] work shows how a cabled system changes into wireless sensor networks and further studies the challenges faced for energy savings and reliable data transmission across the wireless sensor network. Different topologies are tested and compared to data transmission loss and energy efficiency. Finally, tree topology for WSN is adopted to decrease packet loss rate and improve the reliability of data transmission. Reorganising data frames and sleeping of sensor nodes are the two techniques used for Energy saving in this work.

In [22] a system is developed to maintain aquariums properly with ensured WQ. This system uses sensors and actuators to drive the actuators in real-time. Thus the fish is neither over nor underfed, and at the same time, the fish owner can enjoy watching fish feeding remotely. Another work [23] explains the implementation of a robotic device to monitor WQ and feed crayfish. The robotic device consists of a mobile robot

to hold a multiparameter sensor and six food delivery containers. These works explain the working of a WQ monitoring system for fish farming.

This [24] is a case study of a simple WQM system using open-source hardware (Arduino board) and sensors for WQM in aquaculture. The work demonstrates WSN technology in a distributed monitoring system with Arduino as the communication coordinator. The sensors send the information using the Xbee transmitter to the cloud with Arduino or computer acting as the gateway. The Arduino can decide the time to collect information from each sensor and coordinates transmission through the Xbee module. The information is stored and uploaded to the cloud using web service. The mobile application can take data from the cloud using web service for displaying the data.

Another work [25] introduced a novel low-cost compact and robust autonomous buoy system. Commercial WQM systems are costly and unaffordable for small organizations. This work describes the low-cost WQM system fit for use within a near-shore aquaculture farm or bathing waters. The system includes sensors for detecting variations in near-shore physical, chemical and biological WQ. The design is in such a way that this can be easily used by non-trained staff, and provide quality data. The system collected data from two deployments for fourteen months, one in an estuary and other in a coastal bay. These tests demonstrated the robustness of the system and suitability of these sites for aquaculture.

The comparison of the acquired sensor data and independent measurements showed good agreement between them. The paper [26] shows another application of WSN. The focus is on environmental monitoring using a system based on Raspberry Pi and Arduino. The architecture has sensing nodes which act as WSN and uses the ZigBee protocol for data transmission. The base station coordinates the sensor nodes, collects information and updates the database. This information is then uploaded to the internet and can be accessed using cell phones and computers.

This paper [27] provides a real-time WQM system. The Dong Lake in National Dong Hwa University was chosen as the test field to monitor water temperature, turbidity, conductivity, and pH. Various WQ data is updated to the database using wireless transmission, and the website displays the same information. For the long-range data transmission in the field, it uses LoRaWAN (Long Range Wide Area Network) [28]. Also, to extend system life, they use a solar panel with a rechargeable battery.

This [9] presents an online WQM system for intensive fish culture in China, and it has combined web-server-embedded technology with telecommunication. They base it on historical data to design the system to forecast WQ with artificial neural network

(ANN) and control the WQ in time to reduce catastrophic losses. The model forecast DO for half an hour ahead, and then results were compared with experimental data. The results show that WQ information can be accurately acquired and predicted by using this established monitoring system. A work [29] have used narrowband Internet of Things(NB-IoT) for mealtime monitoring of WQ parameters. The system has stable operation, and data transmission is done real-time with accuracy. Another work [30] implements a WQM system for eel fish farming. In this system, sensors measure DO, pH, and temperature, and all data processing is done on a Raspberry Pi and also uses python for all programming. They have automated aeration in this system.

The existing systems were extensively studied. All these works focus on solving the problems with WQ monitoring and not in reducing the cost of the system for making it affordable to small scale farms. This [10] work focuses on reducing the cost by using low-cost sensors. Reducing the cost by using low-quality sensors will not help with the overall cost of monitoring the WQ as the farmers will have to replace the sensors periodically. This [11] studies the available sensors for WQ monitoring in detail. The performance of sensors for each parameter is studied and compared. The challenges of using various types of sensors were also studied.

Research work in [12] presents a WQM system for drinking water. Thorough research identifies low-cost sensors with reliability. These works on the reduction of cost gave direction for reduction of cost of our system. IoT is one of the rapidly rising fields for social and economic development [31]. The field of IoT is extending its wings in all the fields. In this paper [32], the author is studying progression in integrated on-chip open-source computers like Arduino, Raspberry-Pi. These technologies are reaching people with their utilisation in agriculture and aquaculture. They continuously monitor the WQ with sensors to ensure growth and survival and transfer the data to the farmer's mobile through the cloud. They are hence ensuring on-time precautionary steps to reduce the losses and boost productivity.

WQM is an essential aspect of environmental monitoring and control. Traditionally, WQM systems rely on physical sensors that can be expensive and require regular maintenance. In recent years, there has been an increasing interest in developing virtual sensing techniques that can provide accurate and reliable estimates of WQ parameters using data from low-cost, low-power sensors. Virtual sensing is an emerging concept in WQ monitoring that involves the use of computational algorithms to estimate WQ parameters based on other measurable parameters. This approach reduces the need for physical sensors, which can be costly and require frequent calibration.

In [33], authors have developed a DL based virtual soil moisture sensor for pre-

cision farming applications. The authors use LSTM networks to develop the virtual sensor, which is based on data acquired from other environmental sensors installed on the node. The performance of the proposed virtual sensor is compared to other learning-based approaches, and the results show that the proposed method is effective in estimating soil moisture.

Another work [34] presents a data-driven model based on deep dynamic features extracting and transferring methods for virtual sensing of free calcium oxide (f-CaO) content in clinker, which significantly determines the quality of final cement production. The proposed model uses large-scale unlabeled data collected from the process distributed control system (DCS) for feature extraction and a powerful regression model, the eXtreme Gradient Boosting (XGBoost), for output prediction. The proposed virtual sensor is verified by a dataset acquired from a real cement production process.

In [35], the authors propose a magnetoresistive sensor error compensation method using geometry-constraint contour scaling to improve the accuracy of sensitive geomagnetic signals. The proposed method uses a benchmark ellipsoid to determine the shortest distance between any measured single point and the benchmark ellipsoid. The calibration parameters are extracted to complete the single-point error compensation. Simulation and experiment results show that the compensation accuracy of the proposed method is improved by two orders of magnitude compared with the traditional method. This work shows the possibility of using an existing method to improve accuracy of another existing model or method.

In their work [36], the authors present the development of a novel low-cost sensor node for measuring air pollution in real-time using electrochemical and infrared sensors. The sensor node utilizes wireless communication systems such as LoRaWAN and Wi-Fi for long-range, low power communication. The authors also implemented a calibration method using multiple linear regression and ANN, which resulted in significant improvement in the accuracy of the sensor data. This work demonstrates the potential of using low-cost sensor technology and wireless communication systems for air quality monitoring in urban environments.

In [37], authors propose a concept of virtual sensing for hydrogen sulfide (H_2S) gas detection that can augment a single sensing platform into a virtual array, thus increasing the detection accuracy. The authors use a new feature extraction method based on fractional derivatives to enhance the model's learning capabilities. An array of four gas sensors is fabricated to record and analyse the signature of H_2S at various concentration levels, and a shallow neural network (NN) model is trained and tested

on the recorded data. The proposed model is highly sensitive and reliable with the ability to detect the gas instantaneously.

In [38], the authors propose a novel methodology for solving the problem of a water distribution network contamination event, which includes determining the exact source of contamination, the contamination start and end times, and the injected contaminant concentration. The methodology is based on coupling a ML algorithm for predicting the most probable contamination sources in a water distribution network with an optimization algorithm for determining the values of contamination start time, end time, and injected contaminant concentration for each predicted node separately. Two slightly different algorithmic frameworks were constructed, which utilize the Random Forest algorithm for classification of top source contamination node candidates. Both algorithmic frameworks perform well and show robustness in determining the true source node, start and end times, and contaminant concentration.

In summary, the literature survey shows that virtual sensing techniques based on sensors and ML algorithms can provide low cost solutions for aquaculture WQM.

2.2 WATER QUALITY PREDICTION IN AQUACULTURE

Aquaculture is an integral part of the agricultural economy and has undeniably established its crucial role in global food security [39]. Intelligent and precise aquaculture is essential in improving sustainability and reducing the carbon footprint of food production. In intensive aquaculture, WQ is one of the critical factors affecting fish production. The aquaculture WQ is influenced by many factors, such as pollution, climate, quality of fish feed, feeding interval, fish density, etc. Due to the extreme differences in WQ, bacterial and algae balances may be disrupted. This makes WQ a significant factor in affecting aquaculture production. Many aquaculture farms have established WQM systems to monitor WQ in real-time. This approach has helped in managing the WQ in aquaculture to some extent. Further improvement in WQ control can only be realized through prediction, allowing users to take action ahead of time to minimize the occurrence of adverse events. Therefore, aquaculture WQP research is fundamental and has always been crucial in managing the aquaculture WQ. A precise and real-time WQP is necessary to help farmers take measures required to regulate the WQ to make it suitable for high-density aquaculture. Also, with an accurate forecast and the appropriate WQ management, the usage of antibiotics and other chemicals can be minimized, making high-intensity precision aquaculture green, sustainable, and much more profitable [40, 17].

WQM and WQP play a critical part in mitigating the environmental effects of aquaculture production. The WQ parameters need to be maintained in the optimum condi-

tion to ensure healthy fish growth giving good yield, leading to economic benefits [5]. Traditional techniques require collecting samples and transporting them to the lab for further analysis or using test kits. These methods are inaccurate, consume time, require human resources, and make continuous monitoring impossible. However, WQ parameters can be monitored continuously using sensor-based systems, which have advantages like remote 24 x 7 monitoring, control, and real-time data availability. These advantages can incorporate data analysis and application of ML to find optimal WQ parameters for fish growth, optimal use of the resources and predicting the WQ parameters. Thus, maintaining the aquaculture WQ parameters in the optimum range and reducing the impact of aquaculture on the environment.

WQ is the most influential and critical factor that affects production as well as product quality in aquaculture. The vital WQ parameters are salinity, pH, DO, and temperature [4]. Aquaculture species are prone to variations in the WQ parameters. High and unexpected variations in DO, temperature, salinity, pH and other factors can directly result in the mortality of aquaculture species. A slight fluctuation from the ideal environment may trigger stress such as lack of food intake and vulnerability to infections [6, 41]. WQM and WQP are essential to maintain and manage aquaculture [15]. WQM and WQP allow preemptive measures, averting ecological imbalances and ensuring optimal conditions for fish growth. By doing so, aquaculture can witness enhanced productivity, leading to increased profitability and sustainability.

The WQP models earlier used are known as mechanism models, which consider physical, chemical, and biological factors impacting the WQ. The QUAL and QUAL2E based models are two types of mechanism models [42]. But they require an extensive collection of data, including hydrological and meteorological data. The ML-based models are the non-mechanism model that uses past data to predict future data. The classical time series forecasting has methods like ARMA, ARIMA, SARIMA, SARIMAX, HWES, Markov model, grey model, SVR [43–45]. But they have drawbacks such as insufficient accuracy, low generalization, and high computation time. These models assume linear interrelationships between the parameters in WQ data. They cannot handle data with non-linear relation between parameters like aquaculture WQ parameters [18].

The variations in aquaculture WQ parameters are non-linear with respect to time. This non-linear nature of aquaculture WQ makes accurate WQP by using statistical and shallow ML methods a challenging task. There are many conventional time series models for WQP, including auto regressive models and statistical models. [46–50]. Statistical models have strong predictive abilities for stationary time series but need

larger datasets [51–54]. However, all these models have some drawbacks in non-linear prediction, such as lower predictive accuracy, weak generalization capability, and increased time complexity. Hence, these models are not suitable for high-intensity aquaculture WQP [55].

The non-linear nature of the aquaculture WQ parameters has to be handled differently from the mechanical and statistical models. Recently several AI-based models have studied the forecasting of WQ parameters. The authors in [56] have used the SVR model, and the multiple linear regression model (MLRM) for the predicting river water temperature in a river in Karnataka, India. In another work, the authors have used SVR optimised using genetic algorithm to predict the DO and water-temperature [57]. ANN and adaptive neuro-fuzzy inference system (ANFIS), along with its wavelet models are used to predict salinity concentration of a river in Iran by the authors [58]. In another paper, Extreme Learning machine (ELM) models and their improved versions were applied for predicting DO [59]. The ANN, SVR and ELM based models showed promising results in WQP, but there were shortcomings. These models also exhibited poor learning of the complex non-linearity of WQ parameters.

Aquaculture WQ is affected by meteorological conditions and complex interdependency relations between different WQ parameters. Hence the change of WQ parameters exhibit non-linear characteristics, which results in low prediction accuracy. The time-series data shows periodic variations depending on the seasons and climatic conditions. Time series data is used for analysis and prediction in various fields such as the stock market, medical field, energy consumption, weather forecasting, solar radiation etc. [60–65]. A WQP model could describe the relationship between monitored data and trends in WQ parameters based on available data gathered by the WQM system and forecast changes in WQ for a specified period. There are also many shortcomings in conventional prediction approaches, such as weak forecasting precision, least stability etc. Advancements in computer processing power and efficiency have created new prediction models focused on artificial intelligence (AI). Traditional methods for estimating WQ mainly include: the Markov model based on fuzzy-set theory [66, 67], the Grey Markov chain model method [68, 69], the method of prediction of regression [70, 71], the method of time series [72, 73], methods of WQP model [74, 75], the ANN methods [76, 77], the least square support vector machine (LS-SVM) prediction method [78, 79], the combination prediction method [80]. These approaches offer an efficient method for forecasting WQ but are not suitable for the aquaculture environment. The aquaculture environment is affected by many factors such as complex inter-relationship between WQ parameters, non-linearity in the WQ changes etc. Hence these models result in poor prediction accuracy.

Recent advancements in artificial intelligence have made DL one of the most proficient methods, and DL is popularly applied in different fields. DL models consider the non-linear characteristics of WQ changes. DL can determine the essential features that can efficiently provide concise and reliable results [81]. The LSTM model is a widely used DL method for WQP for its remarkable performance in time series prediction [82, 83]. LSTM is an enhanced version of the recurrent neural network (RNN) [84]. Some reported research works using the LSTM model resolves the issues in conventional prediction methods. The authors have used LSTM to predict drinking WQ in [85] and in [86] for mariculture WQP. To solve the problems in aquaculture WQP mentioned above, along with LSTM [87] based DL-RNN, preprocessing and correlation analysis is applied to the data. Correlation analysis uses Pearson's correlation coefficient method. The LSTM based model is used to train and predict WQ parameters. To further enhance the performance, an efficient stochastic optimisation method [88, 89] is used. Adam combines advantage of AdaGrad [90] and RMSProp [91], which are two popular methods. Therefore an Adam optimised LSTM based accurate WQP system for aquaculture is designed.

DL-RNN models have received significant attention due to their ability to capture non-linear inter-dependencies. LSTM, GRU, and CNN models are the DL-RNN models with the ability to capture the non-linearity in the aquaculture WQ. More recently, DL has been widely applied to sequence data processing, time series prediction, and various fields involving the processing of text, image, and speech [20, 92–96]. Even though LSTM and GRU DL-RNN models have overcome the limitations of simple RNN models, they have issues maintaining long-term memory. The forgetting mechanism results in the exponential decay of information, which limit their capacity to capture long-timescale information [97–99]. Past information has the most influence on the prediction at the present step. In certain circumstances, the observation that had a substantial impact may have come long before the current time step. To store memory for the learned long-term dependencies, NNs must be able to keep memories for an extended period. According to recent research [100, 101], the capacity of LSTM models to extract long-term dependency data from past observations remains a crucial performance barrier. The study reported in [97] has theoretically shown that the typical LSTM does not have a long memory. Therefore, a critical issue that must be solved is creating a time-series prediction model capable of collecting and remembering these complex dependencies.

The bidirectional LSTM and GRU (Bi-LSTM and Bi-GRU) models process the data in both ways, as opposed to the standard DL-RNN models, which only process the data in one direction. Therefore, by conducting bidirectional non-linear mapping of the sequence, Bi-LSTM, and Bi-GRU accurately capture the sequence correlation

of data to provide more precise prediction results [102]. Additionally, Bi-DL-RNN models are prevalent in many domains like bio-medical, financial forecasting, electricity consumption and production, weather forecasting, solar radiation, air quality, etc and even for depression identification [102–109]. The use of deep Q-networks in solving optimization problems has been explored in previous research, such as in the case of the flexible job shop scheduling problem and multiobjective FJSP-CS [110]. These studies have demonstrated the effectiveness of deep Q-network (DQNs) [111] in improving efficiency in production and optimizing various objectives. Similarly, the application of DQNs in predicting WQ has shown promising results, indicating its potential in enhancing the accuracy and speed of WQ monitoring and management.

The DL models like LSTM, GRU, and CNN have the flexibility in capturing the non-linear nature of the aquaculture WQP [19]. These DL models have overcome the limitations of RNNs and achieved great success in various applications. Recent advancements in artificial intelligence have made DL one of the most proficient methods, and DL is popularly applied in different fields such as image processing, speech recognition, text prediction, etc. [94–96]. DL makes time series prediction more precise and efficient in terms of training time and required processing power. In addition, DL models consider the non-linear characteristics of WQ changes. LSTM is the widely used DL method for WQP due to its remarkable performance in time series prediction [20].

LSTM and GRU models are not very capable of maintaining long-term memory for time series prediction, especially for long sequences [97–99]. In time series forecasting, the prediction value at the current time step is impacted mainly by past observations. In some cases, the ones with a significant influence might have appeared at the time step long back from the current one. Hence the neural networks (NNs) must have the prolonged memory holding capacity to hold memory for the learned long-term dependencies. The ability of LSTM models to capture long-term dependencies information from historical observations is still considered a critical performance bottleneck, as indicated in recent studies [100, 101]. The work [97] has theoretically proven that standard LSTM does not have long memory from a statistical perspective. Hence, building a time-series prediction model capable of capturing and remembering complex dependencies is a crucial problem that needs to be addressed. Their inability to retain information from distant past steps hampers accurate long-term predictions.

In [99, 112], the authors have discussed how incorporating an attention module can improve the prediction accuracy of LSTM. Different works have combined the attention model to LSTM to capture the relevant data from long time series data, helping to improve its prediction performance. LSTM network with an attention mechanism has an adaptive decay rate of long-term memory. This decay rate is much lower than

the polynomial or exponential decay rate. In the work [113], the reading of the geo-sensor is predicted for a few hours by using multi-level attention-based RNN by feeding data from multiple sensors and meteorological data along with spatial information of these sensors. This work uses the attention mechanism to model the dynamic spatio-temporal relations within the sensors. A fusion module is designed to input the effects of external factors from different domains. The authors have tested their model on a WQ and air quality dataset.

The hybrid CNN-LSTM models also work similarly, helping the LSTM model capture the relevant data. In hybrid models, CNN with maxpooling is used to reduce the length of the input sequence, which is fed into the LSTM and GRU models. The hybrid models are not affected by exponential decay, as the CNN with maxpooling will input the key features and helps LSTM in forecasting with better accuracy. Thus the hybrid models can achieve higher accuracy in prediction. Hybrid models aim to build neither overfit nor underfit models with strong generalization ability and require less training time for fitting the models. The convolutional layer with maxpooling enhances the capacity of the LSTM and GRU models to learn and store the critical features in the non-linear WQ parameters. Thus convolution layer with maxpooling reduces exponential decay in LSTM and GRU, bypassing selective information to the LSTM/GRU layer using convolution and maxpooling [92, 114]. CNNs are faster by design since the computations in CNNs can happen in parallel, while RNNs need to be processed sequentially since the subsequent steps depend on previous ones. In [115], authors introduced the Quasi-Recurrent NNs that use some of the CNN components to imitate RNNs while speeding them up. CNN leads to various complexity reductions by focusing on key features, which consequently reduces the size of tensors. The incorporation of Maxpooling contributes to an additional decrease in training time. As a result, the training time of the hybrid models is effectively reduced.

CNN and its hybrid variations are used in many research work for time series prediction. DL models have been extensively used in many fields of time series prediction, including finance (stock price, cryptocurrency, and precious metal prediction) [116, 117]. The authors [118] have used the CNN-LSTM model for gold price prediction. They have proposed two simple hybrid models and compared the results with other DL models. The authors also compared LSTM models with the hybrid models and observed that hybrid models perform better. In [119] to predict blood glucose (BG) levels, authors have used multiple layers of CNN along with LSTM. This hybrid model has shown superior performance in BG prediction. In another work, the authors have used Bi-LSTM with CNN for air quality prediction [120]. They have done an extensive study with two datasets and shown results proving the model's capability for accurate air quality prediction. There have

been multiple works based on hybrid DL models for the prediction of air quality (PM_{2.5}) based on CNN and LSTM models [121, 122].

The authors of [112] and [123] have described the impact of adding an attention module to increase the performance of the LSTM DL-RNN model. Various studies have used the attention-based LSTM DL-RNN model to extract the relevant information from large time-series data, enhancing prediction accuracy. An attention-based LSTM network features an adaptive long-term memory decay rate. This is much less than the memory's exponential decay rate in the LSTM DL-RNN model. The attention mechanism reduces exponential decay in LSTM and GRU by passing selective information to the LSTM/GRU layer. The attention mechanism effectively highlights the impact of the key features in the prediction model by assigning different weights to hidden layer elements of the NN. Attention mechanisms selectively focus on and extract the essential features of the input data, thus improving model performance.

Attention-based DL-RNN models have also been used for time-series prediction. The authors of the work [124] want to create a unified model to correct the imbalance and enhance the predictability of specific times. The authors have proposed a self-attention-based time-varying prediction model with a two-stage training method [124]. Extensive testing demonstrates that this model has better performance than the baseline DL-RNN models. In [125], authors suggest a versatile and practical DL-RNN model for predicting air pollution. To create the forecasting model based on the novel integrated multiple directed attention (IMDA) DL architecture, the authors first developed the variational autoencoder (VAE) and the attention mechanism. Results show that this model outperforms other DL models, including VAE, LSTM, GRU, Bi-LSTM, Bi-GRU, and ConvLSTM. Results also show significant improvement in the performance of air pollution prediction. In [126], authors suggest SAIFlu-Net, a self-attention-based forecasting model for regional influenza forecasting. The model uses a self-attention mechanism to identify similarities between the occurrence patterns and an LSTM network to extract time-series patterns for each location. The detailed comparisons of the proposed model to other forecasting models utilizing weekly regional influenza datasets reveal that it outperforms other baseline DL-RNN models.

In order to meet the requirement of high prediction accuracy for high-intensity aquaculture, attention-based bidirectional DL-RNN (ABD-RNN) models are proposed in this work. Bi-LSTM and Bi-GRU incorporate additional training layers that allow input data to be trained twice to improve the accuracy of predictions. In this work, the prediction accuracy and training time of the proposed attention-based LSTM and attention-based GRU (ABi-LSTM and ABi-GRU) models are compared

with LSTM, GRU, Bi-LSTM, Bi-GRU, attention-based LSTM (A-LSTM) and attention-based GRU (A-GRU) DL-RNN models for WQP in aquaculture.

Table 2.1: Water Quality Monitoring Approaches in Aquaculture

Approach	Results	Strengths	Weaknesses
Sensor-Based Monitoring System [8]	Effective monitoring of WQ parameters and water level, reducing mortality risk	Real-time monitoring and comprehensive 6-month study	Dependency on battery life and communication performance
Wireless Sensor Networks [21]	Improved reliability of data transmission in aquaculture systems	Tree topology decreases packet loss and energy-efficient design	Challenges in energy saving and data transmission reliability
Aquarium Maintenance System [22]	Real-time management of aquariums, ensuring proper WQ	Automated feeding and remote fish watching capability	Limited to aquarium settings, not scalable for larger aquaculture
Robotic Device for WQ Monitoring [23]	Effective monitoring and feeding in crayfish farming	Integrated multiparameter sensor and food delivery system	High complexity and cost for small scale applications
Open-Source Hardware System [24]	Distributed WQM using Arduino and sensors	Affordable, Scalable and cloud integration	Relies on stable internet connectivity and power supply
Autonomous Buoy System [25]	Long-term data collection in near-shore environments	Low-cost robust design and easy for non-trained staff	Limited to specific environmental conditions
Raspberry Pi Based System [26]	Environmental monitoring with Raspberry Pi and Arduino	Versatile, updatable system and internet connectivity	Requires technical expertise for setup and maintenance
LoRaWAN WQM System [27]	Real-time monitoring of various WQ parameters	Long-range wireless transmission and solar-powered	Limited to areas with LoRaWAN coverage
Online WQM for Intensive Fish Culture [9]	Real-time WQ monitoring and DO prediction	Accurate WQ acquisition with ANN-based forecasting	High initial setup cost, requires historical data
NB-IoT for WQM [29]	Real-time and accurate WQ parameter monitoring	Stable operation and real-time data transmission	Dependency on IoT network availability
Low-Cost Sensor Monitoring [10]	Cost-effective WQM for small scale farms	Reduced costs with low-quality sensors	Frequent sensor replacement, reduced overall reliability

Table 2.2: Water Quality Prediction Approaches in Aquaculture

Approach	Results	Strengths	Weaknesses
SVR and MLRM Models [56]	Prediction of river water temperature	Suitable for specific parameters like temperature	Limited in handling complex interdependencies in WQ parameters
Genetic Algorithm Optimized SVR [57]	Predicts DO and water temperature	Enhanced accuracy through optimization	Requires genetic algorithm knowledge for effective implementation
ANN and ANFIS Models [58]	Prediction of salinity concentration in a river	Handles non-linear data effectively	Complex model structures, requiring substantial computational resources
ELM Models [59]	Prediction of Dissolved Oxygen	Improved prediction accuracy	May not generalize well to different types of WQ parameters
LSTM Models [85, 86]	Prediction of drinking and mariculture water quality	Effective in time series prediction	Challenges in maintaining long-term memory
Adam Optimized LSTM [88, 89]	Accurate WQP system for aquaculture	Combines advantages of AdaGrad and RMSProp	Complexity in implementation and optimization
Bidirectional LSTM and GRU Models [102]	Accurate sequence correlation capture for WQ prediction	Processes data in both directions for precision	Higher computational demands compared to unidirectional models
Hybrid CNN-LSTM Models [92, 114]	Enhanced time series prediction accuracy	Reduction in training time, focusing on key features	May require fine-tuning to balance CNN and LSTM components
Attention-Based LSTM Models [99, 112]	Improved prediction accuracy with attention mechanism	Adaptive decay rate of long-term memory	Increased model complexity and computational needs
Self-Attention-Based Time-Varying Prediction Model [124]	Enhanced predictability for specific times	Addresses imbalance in conventional models	May not be suitable for all types of WQ prediction scenarios

The research in this thesis effectively addresses the key challenges in WQM and WQP, as identified in previous studies through a literature survey, as presented in Tables 2.1 and 2.2. Traditional WQM methods are costly and labor-intensive, unsuitable for small-scale farms, and existing WQP approaches often fail to accurately predict

complex, non-linear water quality parameters. This thesis introduces a Deep Learning Enhanced Virtual Sensing (DLEVS) system for WQM, offering a cost-effective and accurate solution by reducing reliance on expensive physical sensors. For WQP, it utilizes advanced DL models such as LSTM, GRU, their bidirectional variants, and CNN to precisely capture non-linear water quality relationships. The research further innovates with hybrid DL models and attention-based DL models, enhancing prediction accuracy and computational efficiency. This strategic approach not only addresses the limitations of past methodologies but also significantly advances aquaculture technology, promoting sustainable and efficient management practices.

2.3 DATASETS

2.3.1 ADAK water quality dataset

The first dataset used in this research work is the aquaculture WQ data collected from aquaculture farms under ADAK, Kerala. The dataset includes WQ parameters such as salinity, pH, DO, and temperature. As part of this work, the data underwent preprocessing and cleansing. Spanning a three-year period from January 2016 to December 2018, this dataset consists of daily observations collected at a consistent time, resulting in a total of 1096 samples. In terms of size, it is considered a medium-sized dataset when compared to the MAC dataset employed in this research.

2.3.2 MAC water quality dataset

The second dataset is sourced from a publicly available research work [17]. This dataset is acquired from the marine aquaculture base located in Xincun Town, LingShui County, Hainan Province, China. Like the previous dataset, it includes essential WQ parameters such as salinity, pH, DO, and temperature. Data collection occurs at 5-minute intervals. In total, this dataset comprises 23,200 samples gathered over a span of 80 days. In comparison to the ADAK dataset, this is classified as a large dataset.

2.4 PERFORMANCE METRICS

The performance of the prediction models are evaluated using mean absolute error (MAE), mean square error (MSE), root mean squared error (RMSE) and mean absolute

percentage error (MAPE), computed by the set of Eq. (2.1) given below:

$$\begin{aligned}
MAE &= \frac{1}{n} \sum_{i=1}^n |A_i - Y_i| \\
MSE &= \frac{1}{n} \sum_{i=1}^n (A_i - Y_i)^2 \\
RMSE &= \sqrt{\frac{1}{n} \sum_{i=1}^n (A_i - Y_i)^2} \\
MAPE &= \frac{1}{n} \sum_{i=1}^n \frac{|(A_i - Y_i)|}{A_i}
\end{aligned} \tag{2.1}$$

where A_i is the actual value of i_{th} sample, Y_i is the predicted value of i_{th} sample and n is the number of samples.

The chosen metrics, namely MAE, MSE, RMSE, and MAPE, indeed provide different interpretations of model performance. MAE and MSE are measures of average error, with MSE placing greater emphasis on larger errors due to its squared term. RMSE, being the square root of MSE, maintains this emphasis but in the same unit as the data, making it more interpretable. MAPE, expressed as a percentage, offers a scale-independent perspective, making it particularly useful for comparative analysis across different datasets or models.

Regarding their ability to represent both global and local performance, these metrics primarily provide a global summary of the model's performance. However, they can indirectly indicate local discrepancies. A high MSE, for example, may suggest not only an overall inaccuracy but also the presence of significant errors at specific points in the time series.

2.5 LOSS FUNCTION

The primary loss function utilized in this study was the MSE, a widely adopted metric for regression problems, including those within the domain of water quality prediction. MSE proves effective in emphasizing and penalizing larger prediction errors more significantly than smaller ones. This characteristic ensures that models exhibit heightened sensitivity to substantial deviations in predictions, a critical aspect for achieving high accuracy in forecasting water quality parameters.

The selection of MSE as the loss function aligns with the objective of minimizing large errors, thereby enhancing the model's predictive accuracy and reliability. MSE quantifies the average of the squares of the errors, representing the average squared difference between the estimated values and the actual value. This feature makes MSE particularly suitable for models where mitigating large errors is of utmost importance.